The concrete question concerns a very common kind of map in algebraic geometry, and it answers it with one concrete computation. The map takes a space X down to a smaller space Z; the fibres have trivial canonical class in a suitable sense; and the question is how badly the base piece of the canonical bundle formula can be written when you are allowed to fiddle with the representative. The answer says: not badly at all, as long as the total dimension is at most four and the coefficients at the boundary are drawn from a fixed finite set. Every quantity that the formula produces then has a denominator bounded by a constant that knows only the dimension and the coefficient set, never the individual map.

There are three intended readers. The first is a graduate student in algebraic geometry who has met the minimal model program and the canonical bundle formula but who has not yet seen this uniformity argument. The second is a strong undergraduate who wants the shape of the argument and is willing to take some of the deep input theorems on faith. The third is a curious high-school reader who wants to understand what is being claimed and why it is hard. Sections **??** and **??**, and the opening words of every later section, are written for the third audience: they say in plain language what is about to be proved before any formula appears. Everything after those opening words is written at the first audience's level.

Two registers run throughout. There is a *plain form*, one to five sentences, that a reader can take in without stopping. There is a *technical form*, which is the actual statement or proof. The two describe the same object. If the technical form is out of reach, the plain form still tells you what the object is and what it is for.

Two features shape the result: it is a uniformity (not existence) theorem, and it uses that the total dimension is at most four. Sections **??** through **??** prove the main theorem; Sections **??** and **??** apply it.

In words. Take a compact space X that maps onto a smaller space Z, and suppose the fibres have "trivial canonical class". Then a classical theorem says that the canonical class of X is the pullback of a class on Z plus a correction that measures how singular the fibres are. That correction has two pieces: a boundary divisor that records which fibres are badly singular, and a "moduli" part that records something finer, the one-parameter family of fibres through each point of Z. The moduli part is only defined up to adding a principal divisor from the base, and adding a principal divisor can change its coefficients by nasty fractions. The question of the argument is whether, for a fixed dimension bound and a fixed set of allowed boundary coefficients, one can always choose the representative so that those fractions have a common, bounded denominator. The answer is yes, and the bound knows nothing about the map.

Technical setting. Let (X, B) be a projective log canonical pair over $\mathbb{C}$ of dimension d, with B an effective boundary whose coefficients lie in a fixed finite set $\Phi \subset [0,1] \cap \mathbb{Q}$. Let

$$f\colon X \longrightarrow Z$$

be a *contraction*: a surjective projective morphism of normal projective varieties with $f_*\mathcal{O}_X = \mathcal{O}_Z$. Assume $\dim Z > 0$ and that

$$K_X + B \sim_{\mathbb{Q}} f^*D$$

for a rational Cartier divisor D on Z. Then the adjoint of the generic fibre is rationally linearly trivial, and (X, B) is a *log Calabi–Yau fibration* over Z.

The canonical bundle formula. On a suitable smooth base model $q\colon W \to Z$ one writes

$$K_X + B \sim_{\mathbb{Q}} f^*(K_Z + B_Z + M_Z),$$

where B_Z is the *discriminant*: at a prime $P \subset Z$ its coefficient is $1 - t_P$, with t_P the log canonical threshold of f^*P over the generic point of P. The remaining piece $\mathbf{M}$ is the *moduli b-divisor*. It is a b-divisor rather than a divisor because it only descends to a divisor M_W on some high

birational model $W \to Z$, and its pullbacks to higher models are declared. When M_W is nef one says $\mathbf{M}$ is *b-nef*; when pM_W is Cartier one says $p\mathbf{M}$ is *b-Cartier*.

The defect is the phrase "on a suitable model". Change D to a different divisor D_Z in the same rational linear class, and M_W changes by a principal divisor that comes from the base. That can change denominators arbitrarily. The point of an *exact presentation* is to freeze the choice of D_Z so that the moduli part has a denominator controlled from outside.

Plain form of the main theorem. Fix the dimension $d \le 4$ and the coefficient set Φ. There is a single integer p, and a DCC set $\mathcal{B}$ of allowed discriminant coefficients, both depending only on d and Φ, with the following property. For every log Calabi–Yau fibration of this dimension and coefficient set there is a corrected principal divisor $(1/p_0)\operatorname{Div}(\psi)$ and a representative D_Z of the base class such that

$$K_X + B + \frac{1}{p_0}\operatorname{Div}(\psi) = f^*D_Z$$

holds as an equality of actual rational divisors, with $D_Z = K_Z + B_Z + M_Z$, with B_Z effective and coefficients in $\mathcal{B}$, with $(Z, B_Z + M_Z)$ generalized log canonical, and with $p\mathbf{M}$ b-Cartier. The integer p_0 divides p and clears the denominators of Φ. The bound p does not depend on the fibration, only on d and Φ.

Technical form.

Theorem 0.1 (Uniform relative denominators)**.** *Fix a finite set $\Phi \subset [0,1] \cap \mathbb{Q}$ and an integer $1 \le d \le 4$. There are positive integers $p_0 \mid p$, with p_0 clearing Φ, and a rational DCC set $\mathcal{B} \subset [0,1]$, both depending only on d and Φ, with the following property. Suppose (X, B) is a projective log canonical pair of dimension d with coefficients in Φ, and*

$$f\colon X \longrightarrow Z, \qquad \dim Z > 0, \qquad K_X + B \sim_{\mathbb{Q}} f^*D,$$

where f is a contraction and D is a rational Cartier divisor. Then there exist $\psi \in \mathbb{C}(X)^*$ and $D_Z \sim_{\mathbb{Q}} D$ such that

$$K_X + B + \frac{1}{p_0}\operatorname{Div}(\psi) = f^*D_Z, \qquad D_Z = K_Z + B_Z + M_Z, \tag{0.1}$$

where B_Z is effective with coefficients in $\mathcal{B}$, the pair $(Z, B_Z + M_Z)$ is generalized log canonical, and $\mathbf{M}$ is b-nef with $p\mathbf{M}$ b-Cartier. If (X, B) is klt then the generalized pair is generalized klt.

Why the two integers are separate. The integer p_0 does two jobs: it lets the principal correction ψ produce an exact pullback, and on the generic fibre it is a trivializing degree for the log Calabi–Yau pair there. The integer p is larger: it clears the moduli trace on a model that determines $\mathbf{M}$. The discriminant is controlled separately, by the DCC set $\mathcal{B}$.

What is genuinely new. Qualitative versions of the formula put the moduli part on a model and make it nef, but the index may depend on the fibration (Ambro, Fujino–Gongyo). The curve case of Floris even shows that, when the generic fibre is a rational curve and the fibre index is allowed to vary, no polynomial bound on the clearing integer can exist. The theorem says that when the *total* dimension is at most four and the boundary coefficients are from a fixed finite set, the index can be chosen uniformly. The source of the uniformity is the index theorem for the reduced special fibre, whose dimension is one less than the total.
https://arxiv.org/pdf/math/0301180

A short glossary follows. Each item is one or two sentences; a reader who already knows the term can skip it. Everything later refers back by item number. Where a term has a long standard definition only the shape is recorded, not the full definition, and point the reader to Kollár–Mori or the Stacks Project.

(V1): A *variety* is an integral separated scheme of finite type over $\mathbb{C}$. Unless stated otherwise all varieties here are normal and, in the global statements, projective. *Normal* means the local rings are normal domains; a normal variety is smooth outside a closed set of codimension at least two. A *resolution* is a proper birational morphism from a smooth variety.

(V2): A *contraction* $f\colon X \to Z$ is a surjective projective morphism of normal projective varieties with $f_*\mathcal{O}_X = \mathcal{O}_Z$. Equivalently, the function field $\mathbb{C}(Z)$ is algebraically closed in $\mathbb{C}(X)$, so the generic fibre is geometrically integral and the fibres are connected.

(V3): A *divisor* on a normal variety is a formal finite $\mathbb{Z}$-linear combination of prime divisors. A $\mathbb{Q}$-*divisor* allows rational coefficients. A divisor is *effective* when all coefficients are nonnegative. A *boundary* is an effective $\mathbb{Q}$-divisor; the *boundary coefficients* are its coefficients.

(V4): Two $\mathbb{Q}$-divisors are *linearly equivalent*, written $\sim$, when their difference is the divisor of a rational function. They are *rationally linearly equivalent*, written $\sim_{\mathbb{Q}}$, when a positive integer multiple of the difference is. They are *numerically equivalent*, written $\equiv$, when they meet every complete curve the same way. A divisor is *big*, *nef*, or *pseudoeffective* when its class has the corresponding positivity: big means it grows like an ample class on high multiples, nef means nonnegative degree on every curve, pseudoeffective means it lies in the closure of the cone of effective classes.

(V5): The *canonical divisor* K_X of a normal variety is the divisor of a rational top form; on a smooth variety it is the determinant of the cotangent bundle. It is defined up to linear equivalence. A *log pair* (X, B) is a normal variety together with a boundary.

(V6): For a prime divisor E over X (that is, on some resolution $g\colon Y \to X$ with E exceptional or strict transform), the *discrepancy* of E with respect to (X, B) is the coefficient of E in $K_Y - g^*(K_X + B)$ when g is chosen so that this pullback makes sense; it is the discrepancy when K_Y and K_X are fixed compatibly. The pair is *log canonical* (lc) when every discrepancy is at least -1, and *klt* when every discrepancy is strictly bigger than -1. The pair is *dlt* when it is lc and the coefficients are at most one and the crossings are simple normal; dlt is a convenient regular representative of an lc pair.

(V7): The pair is *slc*, semi-log-canonical, when it is defined on a possibly nonnormal variety with pure-dimension components, is S_2, is nodal in codimension one, and its normalization with the conductor included is lc. This is the right singular class on a reduced fibre that may be reducible and nonnormal.

(V8): The *log canonical threshold* $\mathrm{lct}(X, B; P)$ of a prime P over a point of Z is the largest $t \geq 0$ such that the pair $(X, B + t\,f^*P)$ is lc over that point. In the canonical bundle formula the discriminant coefficient at P is $1 - t_P$. A threshold is a single rational number, so it is the natural quantity to try to control uniformly.

(V9): A *b-divisor* $\mathbf{M}$ on a normal variety Z is a rule that assigns a $\mathbb{Q}$-divisor $M_{Z'}$ to every birational model $Z' \to Z$, compatibly with pushforward. It is *b-nef* when $M_{Z'}$ is nef on some model and stays nef under pullback, and *b-Cartier* when some integer $pM_{Z'}$ is Cartier on a determining model. Here $\mathbf{M}$ denotes the moduli b-divisor and $p\mathbf{M}$ b-Cartier means pM_W is an honest Cartier divisor on a model W that determines $\mathbf{M}$.

(V10): A *generalized pair* $(Z, B_Z + M_Z)$ consists of a boundary B_Z and the trace M_Z of a b-divisor on a model. Its adjoint is $K_Z + B_Z + M_Z$, and its discrepancies are read off the crepant boundaries on higher models. It is *generalized lc* (or klt) when those discrepancies are at least -1 (or bigger than -1). The crepant boundaries on higher models may have negative coefficients; that is part of the definition, not an error.

(V11): A *DCC set* is a subset of $\mathbb{R}$ that contains no infinite strictly decreasing sequence; an *ACC set* is the reverse. The relevance is that a DCC set of coefficients cannot contain smaller and smaller rationals approaching a limit from above, and an ACC theorem says that a collection of coefficients produced by a geometric process must lie in some DCC set or even a finite set. The argument uses the ascending chain condition for log canonical thresholds, quoted from Hacon–McKernan–Xu.
https://arxiv.org/pdf/1208.4150

(V12): The *index* of a log Calabi–Yau pair (T, B_T) with $K_T + B_T \sim_{\mathbb{Q}} 0$ is the smallest positive integer n with $n(K_T + B_T) \sim 0$. An *index theorem* bounds n in terms of the dimension and a finite set of allowed boundary coefficients. The argument imports the slc index theorem of Jiang–Liu in dimension at most three.
https://arxiv.org/pdf/2002.11928

(V13): For an integral Weil divisor G on a normal variety, the *reflexive divisorial sheaf* $\mathcal{O}_X(G)$ has sections, on an open U, the nonzero rational functions v with $\mathrm{Div}(v)|_U + G|_U \geq 0$. For a rational divisor D, the *rounded system* is the complete system of $\mathcal{O}_X(\lfloor D \rfloor)$: nonzero v with $\mathrm{Div}(v) + D \geq 0$. This needs no Cartier condition. The *field generated by a system* is the subfield of $\mathbb{C}(X)$ generated by ratios of nonzero sections.

(V14): A *cyclic cover* of a smooth curve C at a point c is the normalization of C in $\mathbb{C}(C)(w)$ with w^m a local uniformizer at c. It is totally ramified of degree m over c and étale elsewhere. Pulling back a fibration to a cyclic cover and comparing a rational generator with its Galois translates is the mechanism by which a denominator is read off a character.

(V15): The *residue* of a rational top form with a logarithmic pole along a smooth divisor is the form obtained by contracting with a local generator of the normal bundle. Iterated residues give a form on a reduced normal crossing fibre, and the residues of two generators differ by a scalar determined by the local node. This is the one analytic-flavoured operation the paper uses.

(V16): *Rational connectedness* of a smooth projective variety Y means that any two points are joined by a chain of rational curves. It implies $\mathrm{Pic}^0(Y) = 0$ and, with it, that the Néron–Severi group is finitely generated. The divisor class group $\mathrm{Cl}(Z)$ of a normal variety with a rationally connected resolution is then finitely generated, and its torsion is bounded using Kummer theory.

In words. The proof of the main theorem has two halves. The first half is conceptual: it produces the exact presentation and turns the moduli bound into a statement about one prime at a time. The second half, Sections **??** and **??**, is a single hard computation for that one prime. The whole strategy is: reduce a global claim to a local number, then compute that number once.

Half one, step by step. The argument starts with the generic fibre of f. It is a log Calabi–Yau pair of dimension $d - \dim Z \leq d - 1 \leq 3$, so the known index theorem in low dimension applies to it. That theorem provides a degree p_0 in which the adjoint of the generic fibre is principal, with an explicit rational function. Because the function field extension is regular, a Galois descent (Hilbert's theorem 90) moves that function down to the actual generic fibre, not just its algebraic closure. The result is a function ψ on X whose $(1/p_0)$-divisor corrects $K_X + B$ to be vertical over Z. A short argument in the same spirit then chooses D_Z in the rational class of D so that the corrected adjoint is literally f^*D_Z, with no vertical error left. That is the exact presentation (0.1).

Once the presentation is exact, the qualitative lc-trivial fibration theorem gives the structural facts for free: the discriminant B_Z is effective and the moduli b-divisor $\mathbf{M}$ is b-nef and descends

on a smooth model. What the qualitative theorem does *not* give for free is a uniform Cartier index for $\mathbf{M}$. That is the content of the argument, and it is what remains to be reduced.

Two independent boundings. The discriminant coefficients $1 - t_P$ must fall into a fixed DCC set; this comes from ACC for log canonical thresholds, applied prime by prime, and is purely qualitative and already uniform. The moduli coefficients must have bounded denominators; this is the hard part. On a smooth model $W \to Z$ with $M_W = D_W - K_W - B_W$, a coefficient of M_W at a prime $P \subset W$ equals an integer (from K_W) plus $\operatorname{coeff}_P D_W$ plus $\operatorname{coeff}_P(-B_W)$. If $\alpha = \operatorname{coeff}_P D_W$ and B_W has coefficient $1 - t_P$ at P over the generic point, that coefficient is $\alpha + t_P$, up to an integer. So it suffices to show that one fixed integer p, independent of the fibration and of P, has

$$p(\alpha + t_P) \in \mathbb{Z} \qquad \text{for every prime } P \subset W.$$

This is the target (0.2) below. Everything in Sections **??** and **??** is aimed at it.

Technical form. Put $k = \dim Z$ and let $K = \mathbb{C}(Z)$. The contraction condition makes K algebraically closed in $\mathbb{C}(X)$, so the extension $\mathbb{C}(X)/K$ is regular; in characteristic zero it is separably generated, and the generic fibre X_η is geometrically integral. It is normal by localization from X, and normality is geometric for a finite-type variety over the perfect field K.

Choose a rational top form on Z, and use it together with the form defining K_X to fix a relative top form on X_η. Its divisor is the canonical divisor K_{X_η}. At codimension-one points of X_η, which are smooth over K, this divisor is the coefficientwise localization of K_X, so vertical primes disappear; when $d = k$, $X_\eta = \operatorname{Spec} K$ and the relative form is a nonzero 0-form. Restricting a log resolution and its crepant sub-boundary to the geometric generic fibre exhibits the original geometric generic pair as lc: generic smoothness makes the horizontal strata simple normal crossing there. That pair is projective of dimension $d - k \le 3$, its boundary coefficients lie in Φ, and its adjoint is rationally linearly trivial.

Lemma 0.2 (generic-fibre trivialization)**.** *There is a positive integer p_0, depending only on d and Φ and clearing the denominators of Φ, and a rational function $\psi \in \mathbb{C}(X)^*$ such that*

$$p_0(K_X + B) + \operatorname{Div}(\psi)$$

is vertical over Z.

Plain form. Apply the slc index theorem to the geometric generic fibre in dimension at most three, getting a trivializing degree. Descend the trivialization across the Galois action by taking ratios of conjugates, whose divisors are vertical and hence constant, and clearing the resulting cocycle with Hilbert 90. □

Technical form. The geometric generic fibre is a projective lc pair of dimension at most three with boundary coefficients in Φ and rationally trivial adjoint. The slc index theorem (Jiang–Liu, quoted as input) gives a common integer p_0, depending only on d and the finite coefficient set, such that $p_0(K_{X_{\overline{\eta}}} + B|_{X_{\overline{\eta}}})$ is principal on the geometric generic fibre. Choosing p_0 also to clear Φ gives an integral divisor.

Descend to K. Choose a finite Galois extension L/K over which a principalizing rational function is defined. The quotient of two Galois conjugates of it has zero divisor on the proper geometrically integral normal fibre over L, hence is a nonzero constant. These quotients form a multiplicative Galois cocycle, and Hilbert's theorem 90 rescales the function to be Galois-invariant; being defined over K, it descends to $\mathbb{C}(X)^*$. Thus there is $\psi \in \mathbb{C}(X)^*$ making $p_0(K_X + B) + \operatorname{Div}(\psi)$ vertical. □

Technical form. Choose a divisible by p_0 and $\phi \in \mathbb{C}(X)^*$ with

$$a(K_X + B) + \operatorname{Div}(\phi) = f^*(aD).$$

Such a pair exists because $K_X + B \sim_{\mathbb{Q}} f^*D$ and a clears denominators. The divisor of $\psi^{a/p_0}/\phi$ is vertical. On the proper normal generic fibre this function has neither zeros nor poles, so it belongs to K^*. Writing it as h, set

$$D_Z = D + \frac{1}{a}\operatorname{Div}(h).$$

Substituting gives the first equality of (0.1) as an equality of actual rational divisors. In particular D_Z is rational Cartier and $D_Z \sim_{\mathbb{Q}} D$.

Technical form. Apply the lc-trivial-fibration theorem of Fujino–Gongyo. Besides generic log canonicity and the rational pullback relation it needs a rank-one condition: on a log resolution the modified discrepancy divisor drops the discrepancy -1 terms, and because $0 \le B \le 1$ rounding up leaves, over the generic point of the base, only an effective exceptional divisor; its divisorial sheaf pushes forward to $\mathcal{O}_X$ by normality, and pushing on to Z has rank one. The theorem then makes $\mathbf{M}$ b-nef and gives its descent to a nef rational Cartier divisor on a sufficiently high smooth projective model $W \to Z$. The statement is stable under changing the rational trivialization, since that adds a rational principal b-divisor from the base.

Write

$$D_W = (W \to Z)^* D_Z, \qquad M_W = D_W - K_W - B_W.$$

For a prime $P \subset W$, set $\alpha = \operatorname{coeff}_P D_W$. Since $\operatorname{coeff}_P B_W = 1 - t_P$ and the coefficient of K_W is an integer,

$$\operatorname{coeff}_P M_W = \alpha + t_P - 1 - \operatorname{coeff}_P K_W.$$

Hence it is enough to prove, uniformly in the choice of P and the fibration,

$$p(\alpha + t_P) \in \mathbb{Z}. \tag{0.2}$$

Once (0.2) holds for a single p, the divisor pM_W is an integral divisor on the smooth variety W, hence Cartier, and $p\mathbf{M}$ is b-Cartier.

Technical form. For primes $P \subset Z$, shrink about the generic point of P so that P is Cartier. In a log resolution computing the threshold, only divisors with positive order along the pullback of P matter; discard those whose images are proper subsets of P. Then t_P is an ordinary log canonical threshold of an lc pair with coefficients in Φ against an effective Cartier divisor with positive integral coefficients. The threshold ACC theorem (Hacon–McKernan–Xu) applies, and since thresholds are rational and lie in $[0,1]$, their complements $1 - t_P$ form the required rational DCC set $\mathcal{B} \subset [0,1]$. The singularities of $(Z, B_Z + M_Z)$ are generalized lc (respectively generalized klt) because the crepant sub-pair over Z is sub-lc (respectively sub-klt): $t_P \ge 0$ always, and $t_P > 0$ in the klt case. On the original base Z the boundary upstairs is effective with positive integral multiplicity over P, which forces $t_P \le 1$ and hence $B_Z \ge 0$.

After these considerations the main theorem is reduced to a single statement: there is an integer p, depending only on d and Φ, such that the coarse identity

$$p(\alpha + t_P) \in \mathbb{Z} \qquad \text{for every prime } P \text{ of every determining model } W \tag{0.3}$$

holds. The threshold t_P is disguised: it was computed as an lc threshold over the generic point of P, but the required uniformity is insensitive to the individual P. The next two sections compute (0.3) by slicing to a curve through the generic point of P, running a relative minimal model program there, and reading the answer off a residue on the resulting reduced fibre.

In words. The goal is to prove one congruence, $p(\alpha+t_P) \in \mathbb{Z}$, for one prime P of one determining model W. The trick is to make the situation one-dimensional by cutting with general divisors through a general point of P. A curve has no room for complications: the "fibre" over a point of the curve is a divisor, the singularities are controlled, and a residue computation can see the denominator. So slice and run the relative minimal model program, and reduce the congruence to a statement about a single reduced special fibre.

The plan. Fix $P \subset W$. Build a smooth model $V \to X$ that maps to W and is marked along everything relevant. Cut V by $k-1 = \dim W - 1$ general very ample divisors to get a curve C and a total space $V_C \to C$. On V_C write an exact relative identity whose base coefficient at the chosen point $c \in C$ is exactly α. Construct a dlt pair by adding the reduced fibre and a horizontal boundary. Run a minimal model program over C. The program does not change the function field and does not change the coefficient of interest; it makes the total space terminal enough that the reduced fibre is well behaved. The output is a dlt model $N \to C$ on which an exact divisor relation holds:

$$L + \operatorname{Div}(\psi_C) = p_0(\alpha + t) f_N^*[c], \qquad L = p_0(K_N + T + H),$$

with T the reduced fibre. That relation is handed directly to the residue lemma of Section **??**.

Technical form. Fix a prime $P \subset W$. Take a smooth projective model $V \to X$ mapping to W by $g\colon V \to W$, with simple normal crossing markings for the crepant boundary B_V^c, the exceptional divisors over X, and the support of g^*P. Because the function field extension $\mathbb{C}(W) \subset \mathbb{C}(V)$ is regular (the same argument as for f), the Stein factorisation of g is W itself, so g has connected fibres. Let θ_V be the rational top form compatible with the chosen canonical divisor on X, and put $K_V = \operatorname{Div}(\theta_V)$. Crepant pullback gives

$$K_V + B_V^c + \frac{1}{p_0}\operatorname{Div}(\psi) = g^* D_W. \tag{0.4}$$

Put $t = t_P$. At the generic point of P, the boundary $B_V^c + t\, g^*P$ has coefficients at most one, with equality on at least one component dominating P; this is the threshold computation.

Choose general very ample divisors $A_1, \dots, A_{k-1}$ on W (taking $C = W$ when $k = 1$) and let $C = A_1 \cap \dots \cap A_{k-1}$. Bertini applied successively to the pulled-back systems and the finitely many marked strata makes C and $V_C = g^{-1}C$ smooth, with the markings still SNC. The curve C is connected; the restriction of g has connected fibres, so V_C is connected and hence integral. Also required are $C \not\subset \operatorname{Supp} D_W$ and $V_C \not\subset \operatorname{Supp} K_V \cup \operatorname{Supp} B_V^c \cup \operatorname{Supp}\operatorname{Div}(\psi)$. The dimension of V_C is $d - k + 1 \le 4$. Choose an intersection point $c \in C \cap P$ transverse and general on P, avoiding at c all other components of $\operatorname{Supp} D_W$, the bad strata on P, and the images of all nondominating strata not containing P.

Technical form. Let $h : V_C \to C$, $F = h^*[c]$, and $B^c = B_V^c|_{V_C}$. To keep the coefficient of P the canonical divisor is specified using complete-intersection adjunction. Choose general representatives $A_j' \sim A_j$ avoiding c and rational functions $\chi_j \in \mathbb{C}(W)^*$ with $\operatorname{Div}(\chi_j) = A_j' - A_j$. Define θ_C as the iterated Poincaré residue, in the order of the cuts, of

$$\theta_V \prod_{j=1}^{k-1} g^*\chi_j$$

along $g^{-1}A_1, \dots, g^{-1}A_{k-1}$. Generality makes this a nonzero rational top form on V_C; for $k = 1$ the product is empty and the residue is the identity. Each χ_j supplies a simple pole along its divisor and a zero along its chosen representative, so, cancelling the cutting divisors before restriction,

the residue formula gives an equality of actual divisors

$$K_{V_C} := \operatorname{Div}(\theta_C) = \left(\operatorname{Div}\left(\theta_V \prod_j g^* \chi_j \right) + \sum_j g^* A_j \right) |_{V_C}$$
$$= \left(K_V + \sum_j g^* A'_j \right) |_{V_C}.$$

Set $\psi_C = \psi|_{V_C}$ and $D'_C = (D_W + \sum_j A'_j)|_C$. Restricting (0.4) gives

$$K_{V_C} + B^c + \frac{1}{p_0} \operatorname{Div}(\psi_C) = h^* D'_C, \qquad \operatorname{coeff}_c D'_C = \alpha. \tag{0.5}$$

The coefficient assertion uses that C meets P transversely at c while the A'_j and all other components of $\operatorname{Supp} D_W$ avoid c.

Technical form. Along F, the coefficients of $B^c + tF$ are at most one, with equality on a component of F: the divisors dominating P restrict transversely and reducedly, and near c, $g^*P|_{V_C} = F$. Over the generic point of C there is still a projective birational comparison with the normal fibre of the original model X, and the restricted exceptional divisors are exceptional there; this is because the generic point of C lies where $W \to Z$ is an isomorphism, and generic flatness and openness of geometric normality give normal integral original fibres on an open base with an open birational isomorphism locus whose exceptional images have fibre codimension at least two.

On V_C define an effective boundary Γ as F_{red}, plus the horizontal strict transforms of the original boundary with their original coefficients, plus all horizontal restricted exceptional divisors with coefficient one. Then (V_C, Γ) is log smooth, its horizontal coefficients lie in the fixed set $\Phi \cup \{1\}$, and

$$E = \Gamma - B^c - tF \tag{0.6}$$

is effective near c and has coefficient zero on a component of F. On the generic fibre, E is effective: its horizontal nonexceptional terms cancel, while every horizontal exceptional term has coefficient $1 - \operatorname{coeff} B^c \geq 0$, and its support there is exceptional over the original normal fibre. Equation (0.5) gives the exact identity

$$K_{V_C} + \Gamma + \frac{1}{p_0} \operatorname{Div}(\psi_C) = h^*(D'_C + t[c]) + E.$$

Run a minimal model program over C. All negative coefficients of E are vertical and occur over finitely many points, so there is an effective rational divisor A on C with $E^+ := E + h^*A \geq 0$. The identity becomes

$$K_{V_C} + \Gamma + \frac{1}{p_0} \operatorname{Div}(\psi_C) = h^*(D'_C + t[c] - A) + E^+.$$

In particular $K_{V_C} + \Gamma \sim_{\mathbb{Q},C} E^+$, whose restriction to a very general fibre is rationally linearly equivalent to an effective divisor, hence pseudo-effective over C.

Technical form. The existence-and-preservation result of Chen–Tsakanikas permits a minimal model program with scaling of an ample over the base for an lc generalized pair whose underlying variety is $\mathbb{Q}$-factorial klt; applied to the $\mathbb{Q}$-factorial dlt pair (V_C, Γ) over C, it keeps the stages $\mathbb{Q}$-factorial and dlt, and each step is negative for the transformed adjoint. The zero nef datum satisfies the NQC hypothesis. In dimension four, their theorem terminates every flip sequence over C for an NQC lc generalized pair whose adjoint is pseudo-effective over C; in dimension three their statement has no pseudo-effectivity hypothesis. These give the program and terminate the flip tail.

Fix canonical divisors on every birational stage using θ_C, and push forward Γ, E^+; every step is over C, so the identity pushes forward to

$$K_{V_i} + \Gamma_i + \frac{1}{p_0}\operatorname{Div}(\psi_C) = h_i^*(D'_C + t[c] - A) + E_i^+, \qquad E_i^+ \geq 0.$$

A fibre-type negative extremal contraction is impossible: a general positive-dimensional contracted fibre has a curve ℓ not contained in $\operatorname{Supp} E_i^+$, and $\mathbb{Q}$-factoriality plus effectivity makes the intersection $E_i^+ \cdot \ell \geq 0$, while the contraction makes it negative, a contradiction. Divisorial contractions strictly decrease the relative Picard number, so an infinite program would eventually consist only of flips, which the termination statements exclude; in dimensions at most two no flips occur. The endpoint is a projective morphism $f_N : N \to C$ with N $\mathbb{Q}$-factorial, (N, Γ_N) dlt, and $K_N + \Gamma_N$ nef over C. The unchanged function field and Stein factorisation give connected fibres, so f_N is a contraction.

Technical form. Let E_N^+ be the transform of E^+ on N. The pushforward of h^*A is f_N^*A, so $E_N := E_N^+ - f_N^*A$ is the pushforward of the original E, and the transformed identity reads

$$K_N + \Gamma_N + \frac{1}{p_0}\operatorname{Div}(\psi_C) = f_N^*(D'_C + t[c]) + E_N. \tag{0.7}$$

Take a common resolution $r : \widetilde{V} \to V_C$, $r' : \widetilde{V} \to N$. Discrepancy comparison for a $K + \Gamma$ program gives

$$r^*(K_{V_C} + \Gamma) - r'^*(K_N + \Gamma_N) \geq 0,$$

and cancelling common base pullbacks and principal divisors in the input and output identities gives

$$r'^*E_N \leq r^*E. \tag{0.8}$$

Let η be the generic point of C and $q : \widetilde{V}_\eta \to X_\eta$ the map to the normal original fibre. The restriction of E_N to N_η is effective, being the pushforward of the effective restriction of E; the restriction of r^*E is effective and q-exceptional; so by (0.8) the restriction of r'^*E_N is q-exceptional. But (0.7) gives $E_N \equiv_C K_N + \Gamma_N$, which is nef over C, and hence q-nef. The negativity lemma makes it zero. Thus E_N has no horizontal component; after shrinking about c it is effective, supported on the special fibre, and nef over C.

Write $F_N = f_N^*[c] = \sum_i e_i S_i$ and $E_N = \sum_i q_i S_i$ with $e_i > 0$, $q_i \geq 0$, and set $\lambda = \max_i(q_i/e_i)$. Then $\Delta = \lambda F_N - E_N$ is effective, anti-nef over C, and misses at least one component of the fibre. If the fibre has positive dimension and $\Delta \neq 0$, connectedness gives a component S outside $\operatorname{Supp}\Delta$ meeting it; since N is $\mathbb{Q}$-factorial a positive multiple $b\Delta$ is Cartier, and its restriction to the integral projective S is a nonzero effective Cartier divisor, so $((b\Delta)|_S) \cdot A_S^{n-1} > 0$ for very ample A_S, contradicting anti-nefness. Hence $\Delta = 0$, i.e. $E_N = \lambda F_N$; for a zero-dimensional fibre, f_N is an isomorphism of normal curves and the proportionality is immediate. Since $r'^*F_N = r^*F$ and E has coefficient zero on a component of F, comparing that coefficient forces $\lambda = 0$. So $E_N = 0$ near c.

Technical form. Shrink C about c and set $T = (f_N^*[c])_{\mathrm{red}}$. No divisor was extracted in the program, so every surviving component of this fibre has coefficient one in Γ_N; hence $\Gamma_N = T + H$, with H horizontal and coefficients in $\Phi \cup \{1\}$. The reduced fibre T is connected and projective, and its components have dimension $d - k \leq 3$. Removing the boundary from the $\mathbb{Q}$-factorial dlt pair shows that N is klt. Shrink again so that D'_C is supported at most at c. For the integral Weil divisor $L = p_0(K_N + T + H)$ the exact relation is now

$$L + \operatorname{Div}(\psi_C) = p_0(\alpha + t)f_N^*[c]. \tag{0.9}$$

This is exactly the hypothesis of the residue lemma, with $\beta = \alpha + t$. The remaining work is to show that one uniform p, depending only on d and Φ, has $p\beta \in \mathbb{Z}$ in every such configuration.

In words. This is the one genuinely delicate computation. The setup provides handed a normal total space N over a curve, a reduced fibre T, and a horizontal boundary H, with an exact relation

$$p_0(K_N + T + H) + \mathrm{Div}(\psi) = p_0\beta\, F, \qquad F = f^*[c], \quad T = F_{\mathrm{red}}.$$

The question is whether $p\beta$ is an integer for one fixed p that does not depend on the configuration. The strategy is: restrict the relation to the reduced fibre, where it becomes a statement that a certain log Calabi–Yau pair (T, B_T) has a trivial p-th pluricanonical class; take a global generator u of that trivialization; pull back to a cyclic cover of the curve that opens up the denominator; and show, by comparing residues across the nodes of the cover, that the would-be extra character is forced to be trivial. The denominator must therefore divide p.

Technical form.

Lemma 0.3 (The reduced fibre and the cyclic residue test)**.** *Fix a finite set $\Phi \subset [0,1] \cap \mathbb{Q}$ and a positive integer p_0 clearing its denominators. There is a positive integer p, depending only on Φ and p_0, divisible by p_0, with the following property. Let*

$$f\colon N \longrightarrow C$$

be a projective contraction from a normal integral variety to a smooth complex curve, let $c \in C$, and suppose $1 \le \dim N \le 4$. Set $F = f^*[c]$ and $T = F_{\mathrm{red}}$. Assume that N is $\mathbb{Q}$-factorial, that $(N, T + H)$ is dlt, and that H is an effective horizontal $\mathbb{Q}$-divisor with coefficients in Φ. Suppose, on a neighbourhood of F, that there are $\beta \in \mathbb{Q}$ and $\psi \in \mathbb{C}(N)^*$ such that, for a fixed canonical divisor,

$$p_0(K_N + T + H) + \mathrm{Div}(\psi) = p_0\beta F \tag{0.10}$$

as an equality of actual $\mathbb{Q}$-divisors. Then $p\beta \in \mathbb{Z}$.

Proof. The proof has four movements. Adjoin on the whole reduced fibre to get an slc pair (T, B_T) with a uniform index. Take a global generator u and a cyclic cover that unwinds the denominator. Compare residues on the components of the covering fibre and glue the scalars. Kill the remaining character.

Technical form. Shrink C about c throughout; fix a uniformizer $z \in \mathbb{C}(C)$ with $\mathrm{Div}_C\, z = [c]$ there. The fibre T is connected and projective, and its irreducible components have dimension $\dim N - 1$, because F is an effective Cartier divisor on the integral variety N with support the whole fibre. When $\dim N = 1$, f is an isomorphism of normal curves and comparing the coefficient at c already gives $p_0\beta \in \mathbb{Z}$; so assume $1 \le \dim T \le 3$.

Write $T = \bigcup_i T_i$. By dlt adjunction and the structure of dlt strata, each T_i is normal and

$$(K_N + T + H)|_{T_i} \sim_{\mathbb{Q}} K_{T_i} + \Theta_i, \tag{0.11}$$

where (T_i, Θ_i) is lc and Θ_i is the effective different. If n is sufficiently divisible, the n-th tensor power of the ordinary residue at the generic point of T_i extends to an isomorphism

$$\mathcal{O}_N(n(K_N + T + H))|_{T_i} \simeq \mathcal{O}_{T_i}(n(K_{T_i} + \Theta_i)),$$

identifying actual rational pluricanonical forms.

Intersections of distinct components are unions of lc strata. At the generic point of a codimension-two intersection in N the pair is simple normal crossing, exactly two components of

T pass through the point, and H is absent; the conductor prime on either branch has coefficient one in Θ_i. The remaining coefficients of the differents lie in the set

$$\mathcal{D}(\Phi) = \left\{ \frac{r-1+b}{r} \;\middle|\; r \in \mathbb{N}_{>0},\ b = \sum_{\phi \in \Phi \cup \{1\}} n_\phi \phi \le 1 \right\} \cap [0,1], \tag{0.12}$$

which is a fixed rational DCC set: the possible sums $b \le 1$ form a finite set, and for each such sum the displayed coefficients increase in r toward one.

To apply the index theorem to T itself and not just to its components, T must be the right kind of scheme. Apply the depth theorem to the dlt pair $(N, T+H)$ with the effective integral divisor $D = T \le \lfloor T+H \rfloor$ and $L = 0$; it makes $\mathcal{O}_N$, $\mathcal{O}_N(-T)$, and $\mathcal{O}_T$ Cohen–Macaulay, so T satisfies S_2. The normality of the T_i and the generic crossing description make T smooth away from ordinary double curves, hence seminormal (the square–cube criterion, checked at codimension-one points, plus the S_2 extension property). Thus T is demi-normal, its normalization is the disjoint union of the T_i, and its conductor is the union of the double curves.

Remove the coefficient-one conductor divisors from the Θ_i and push the remaining boundary cycles to T; call the result B_T. Off the conductor the normalization is an isomorphism in codimension one, so nothing is double counted. The boundary B_T is effective, has coefficients in $\mathcal{D}(\Phi)$, and contains no conductor component. Choosing an open $j : T^\circ \hookrightarrow T$ whose complement has codimension at least two and where T° has only smooth points and ordinary double crossings, one defines $\mathcal{O}_T(n(K_T + B_T))$ first on T° and then extends by S_2. For every sufficiently divisible even n the residue maps give a natural isomorphism

$$\mathcal{O}_N\big(n(K_N + T + H)\big)|_T \simeq \mathcal{O}_T\big(n(K_T + B_T)\big). \tag{0.13}$$

At a node with normal-crossing coordinates x, y, hypersurface adjunction for $xy = 0$ matches the ambient log canonical line with the dualizing line of the node; the two iterated residues of $dx/x \wedge dy/y$ differ by a sign whose n-th power is $+1$ for even n, and their next residues agree. It follows that $K_T + B_T$ is $\mathbb{Q}$-Cartier and its pullback to the normalization is $K_{T_i} + \Theta_i$ with the conductor included, so (T, B_T) is slc. Combining (0.10) with $F = \mathrm{Div}(f^* z)$ and restricting a sufficiently divisible trivialization in (0.13) gives

$$K_T + B_T \sim_{\mathbb{Q}} 0. \tag{0.14}$$

Technical form. Apply global ACC to the projective normalized component pairs (T_i, Θ_i), whose adjoints are numerically trivial by (0.14). Their dimensions are at most three and their coefficients lie in the DCC set $\mathcal{D}(\Phi)$, so their nonzero coefficients lie in a fixed finite rational set; likewise for B_T. The slc index theorem in dimension at most three then gives a common integer annihilating $K_T + B_T$. Enlarging it, choose once and for all an even multiple p of p_0, depending only on Φ and p_0, with

$$pB_T \text{ integral}, \qquad p(K_T + B_T) \sim 0. \tag{0.15}$$

Choose a nowhere-vanishing generator u of this trivial log pluricanonical line; on the normalization it is a tuple of rational p-pluricanonical forms u_i with the prescribed boundary poles and the conductor gluing.

Technical form. Write

$$p_0\beta = \frac{a}{m}, \qquad a \in \mathbb{Z},\ m \in \mathbb{N}_{>0},\ \gcd(a,m) = 1. \tag{0.16}$$

The case $m = 1$ is immediate; assume $m > 1$. Let $C' \to C$ be the normalization in $\mathbb{C}(C)(w)$ with $w^m = z$, and let $\pi : Y \to N$ be the normalization of the dominating component of $N \times_C C'$. The polynomial $w^m - z$ is Eisenstein at c, so C' has a unique point c' over c with total ramification of

degree m; and since $\mathbb{C}(C)$ is algebraically closed in $\mathbb{C}(N)$ and the extension is regular, $\mathbb{C}(C')/\mathbb{C}(C)$ is linearly disjoint from it. The map π has degree m and cyclic Galois group μ_m acting by $w \mapsto \zeta w$. Write $f_Y : Y \to C'$ and $T_Y = (\pi^{-1}T)_{\mathrm{red}}$. Figure 1 records the maps; Y is the normalization of the base change.

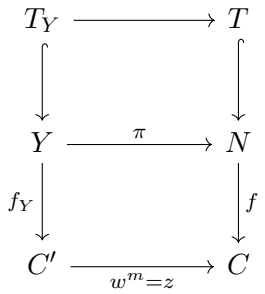


FIGURE 1. The normalized cyclic base change and its whole reduced special fibres over c' and c. The covering group may permute components of T_Y, so the residue comparison must use their conductor gluing.

The morphism f_Y is projective and its function field is regular over $\mathbb{C}(C')$, so Stein factorisation gives $(f_Y)_*\mathcal{O}_Y = \mathcal{O}_{C'}$; in particular T_Y is connected and projective. The divisor $p_0(K_N + T + H)$ in (0.10) is integral, and comparing coefficients with (0.16) shows m divides every coefficient e_i of $F = \sum_i e_i T_i$. At the generic point of T_i write $f^*z = \tau^{e_i} v$ with v a unit; after dividing w by $\tau^{e_i/m}$ the normalized extension is obtained by extracting an m-th root of a unit and is étale. Away from T the function f^*z is a unit and the same holds, so π is étale in codimension one.

Put $H_Y = \pi^*H$ and choose canonical divisors compatibly. Since π is quasi-étale and the coefficient-one divisor pulls back without divisorial ramification,

$$K_Y + T_Y + H_Y = \pi^*(K_N + T + H). \tag{0.17}$$

The pair on the left is dlt. Finite crepant discrepancy comparison multiplies a downstairs log discrepancy by the ramification index of a divisorial prolongation, so the upstairs pair is lc and every upstairs lc centre maps onto a downstairs lc centre; a finite quasi-étale map is étale over the smooth locus of its target by purity of the branch locus, so the upstairs pair is simple normal crossing at the generic points of all its lc centres, which is the dlt condition. In particular each component $T_{Y,j}$ of T_Y is normal.

Technical form. Let θ be the rational top form defining K_N, and define

$$s = (\pi^*\theta)^{\otimes p_0}\, \pi^*\psi\, w^{-a}. \tag{0.18}$$

Since $\pi^*F = m\,\mathrm{Div}_Y\, w$, equations (0.10) and (0.17) give $\mathrm{Div}(s) + p_0(T_Y + H_Y) = 0$. Thus s generates the degree-p_0 divisorial log pluricanonical sheaf on Y, and its sufficiently divisible powers generate the corresponding invertible sheaves.

Technical form. Let T_i be the component of T dominated by $T_{Y,j}$, and let $\rho_j : T_{Y,j} \to T_i$ be the induced finite étale map at their generic points. The generic residue

$$v_j = \mathrm{res}_{T_{Y,j}}\big(s^{p/p_0}\big)$$

is a nonzero rational p-pluricanonical form, as is $\rho_j^* u_i$, and the quotient

$$r_j = \frac{v_j}{\rho_j^* u_i} \in \mathbb{C}(T_{Y,j})^* \tag{0.19}$$

is a rational function. Choose a multiple $n = bp$ sufficiently divisible for (0.13) and for component adjunction upstairs. Then v_j^b is a nowhere-vanishing generator of the degree-n adjunction sheaf on $T_{Y,j}$. The same is true of $\rho_j^* u_i^b$: near any point of T, the section u^b is a generator of the restriction of the invertible sheaf $\mathcal{O}_N(n(K_N + T + H))$, hence is a unit relative to a local ambient generator; such a unit lifts after restricting the neighbourhood, so u^b is the residue restriction of a local ambient log generator. Pull that generator back to Y; it remains a log generator by (0.17), and its residue on $T_{Y,j}$ generates the upstairs adjunction sheaf and equals $\rho_j^* u_i^b$ because residue commutes with étale pullback. Therefore r_j^b, and hence r_j, has zero divisor; a rational function with zero divisor on a normal projective integral variety is a nonzero constant. So

$$v_j = r_j \, \rho_j^* u_i, \qquad r_j \in \mathbb{C}^*. \tag{0.20}$$

Technical form. Suppose two distinct components $T_{Y,j}$ and $T_{Y,j'}$ meet. Their intersection contains a stratum of codimension two in Y along which the pair is generically simple normal crossing, and its image is a codimension-two lc centre downstairs where π is étale and the downstairs pair is simple normal crossing. The two downstairs vertical branches are distinct, and H is absent at this generic crossing. At the generic point ξ of the upstairs crossing choose the two normal coordinates x, y and a regular ambient $(\dim Y - 2)$-form η. Since s^{p/p_0} is a log generator, it has the local form

$$s^{p/p_0} = \varepsilon \left(\frac{dx}{x} \wedge \frac{dy}{y} \wedge \eta \right)^{\otimes p}, \qquad \varepsilon \in \mathcal{O}_{Y,\xi}^*.$$

The two next residues differ by $(-1)^p$ and are nonzero, because ε is a unit; as p is even they agree. The components of u have the same matching, nonzero next residues downstairs, since u generates the slc log pluricanonical line whose local node description is the even tensor power of the dualizing line, and étale pullback preserves this. Taking next residues in (0.20) gives $r_j = r_{j'}$.

Technical form. Since T_Y is connected, its component-incidence graph is connected, so all r_j equal one scalar $r \in \mathbb{C}^*$. In tuple notation on the normalized components of T_Y,

$$\left(\operatorname{res}_{T_{Y,j}}(s^{p/p_0})\right)_j = r \, \pi^* u, \tag{0.21}$$

and $\pi^* u$ is invariant under the covering group, with the natural action also permuting components. On the other hand, for $\zeta \in \mu_m$, (0.18) gives $\zeta^*(s^{p/p_0}) = \zeta^{-ap/p_0} s^{p/p_0}$, and residue is natural under this action including its permutation of components. The nonzero scalar comparison (0.21) forces this character to be trivial. So $m \mid ap/p_0$; since $\gcd(a, m) = 1$, obtain $m \mid p/p_0$, and hence

$$p\beta = \frac{p}{p_0} \cdot \frac{a}{m} \in \mathbb{Z}.$$

This proves the lemma. □

Apply Lemma 0.3 to (0.9) with $\beta = \alpha + t$ and the fixed coefficient set $\Phi \cup \{1\}$. It gives a single multiple p of p_0, depending only on d and Φ, for which (0.2) holds at every prime P of the determination W. Consequently pM_W is integral and Cartier, and since the moduli b-divisor descends on W the b-divisor $p\mathbf{M}$ is b-Cartier. Every construction used only the dimension bound $d \leq 4$ and the fixed coefficient set Φ. Together with the discriminant bound of Section **??** this completes the proof of Theorem 0.1.

In words. The main theorem was about writing the base divisor with bounded denominators. A closely related question is: can pluricanonical sections upstairs actually *see* the base downstairs? When the base divisor is big, the answer is yes in one uniform degree: for every large enough multiple l, the complete rounded system $|\lfloor l(K_X + B) \rfloor|$ is nonempty, and the field generated by

ratios of its sections is exactly the base function field $\mathbb{C}(Z)$, no more and no less. "No more" is the substantive half: the sections must not accidentally generate a larger subfield of $\mathbb{C}(X)$.

Technical form.

Proposition 0.4 (Effective systems recovering the full base field)**.** *Under the hypotheses of Theorem 0.1, suppose in addition that D is big. There is a positive integer $m = m(d, \Phi)$ such that, for every positive multiple l of m, the complete divisorial system*

$$||\lfloor l(K_X + B)\rfloor||$$

is nonempty and generates precisely the subfield $\mathbb{C}(Z) \subset \mathbb{C}(X)$. Its rational map is therefore birationally equivalent to f and to the Iitaka fibration of $K_X + B$.

Plain form. First identify the complete section space upstairs with the pullback of a complete section space on the base, using the exact presentation and a valuation argument valid for reflexive sheaves. Then produce birational sections on the base by the polarized effective-birationality theorem applied to a big adjoint on a smooth model. To identify the Iitaka fibration, observe that multiples of m form a cofinal divisible subsequence of the Cartier section series, which has the same field of ratios. □

Technical form. Put $L = K_X + B$ and $K = \mathbb{C}(Z)$, and use the exact presentation (0.1). For every positive integer l divisible by p_0, the equality of subspaces of $\mathbb{C}(X)$ is first proved

$$H^0(X, \mathcal{O}_X(\lfloor lL\rfloor)) = \psi^{l/p_0} f^* H^0(Z, \mathcal{O}_Z(\lfloor lD_Z\rfloor))\,, \tag{0.22}$$

where f^* on the right means pullback of rational functions. For $v \in K^*$, the exact divisor identity gives

$$lL + \mathrm{Div}_X\left(\psi^{l/p_0} f^* v\right) = f^*(lD_Z + \mathrm{Div}_Z(v))\,. \tag{0.23}$$

If v is a section of the rounded system on Z, then $lD_Z + \mathrm{Div}_Z(v) \geq 0$ by the section convention of Section **??**. This divisor is rational Cartier, so local effective Cartier multiples show that its pullback is effective; (0.23) gives the inclusion from right to left in (0.22).

Conversely, let $u \neq 0$ be a section on the left and let X_η be the generic fibre over K. The exact presentation rewrites the section inequality as $\mathrm{Div}_X(u/\psi^{l/p_0}) + lf^*D_Z \geq 0$, whose horizontal valuations give

$$\mathrm{Div}_{X_\eta}\left(u/\psi^{l/p_0}\right) \geq 0.$$

The generic fibre is normal, so this quotient is regular there; and the contraction condition $f_*\mathcal{O}_X = \mathcal{O}_Z$ gives $H^0(X_\eta, \mathcal{O}_{X_\eta}) = K$. Hence $u = \psi^{l/p_0} f^* v$ for some $v \in K^*$. By (0.23) the source inequality is $f^*G \geq 0$ where $G = lD_Z + \mathrm{Div}_Z(v)$ is rational Cartier. Effectivity of G descends to Z: near the generic point of any prime $P \subset Z$, write a Cartier multiple of G as aP with P locally principal; the local Cartier pullback of P has a prime component dominating P with positive multiplicity e because f is surjective, and its coefficient in the pullback of the chosen multiple is ae. Thus $f^*G \geq 0$ forces $a \geq 0$. Doing this for every P proves $G \geq 0$, so v is a section on Z, and (0.22) holds. Since p_0 clears Φ, the divisor lL is integral Weil, and the argument applies to its reflexive sheaf even when lL is not Cartier.

Now produce birational sections on the base. Choose a smooth projective determination $q : W \to Z$ of the nef data that also resolves the boundary, and write

$$K_W + B_W^{\mathrm{cr}} + M_W = q^* D_Z.$$

Let A be the strict transform of B_Z plus the reduced exceptional divisor. On nonexceptional primes A agrees with B_W^{cr}; on exceptional primes its coefficient is one, at least the coefficient of

B_W^{cr} by generalized log canonicity. Thus

$$K_W + A + M_W = q^*D_Z + E_W, \qquad E_W = A - B_W^{\mathrm{cr}} \geq 0 \text{ exceptional over } Z. \tag{0.24}$$

Put $\Lambda = \mathcal{B} \cup \{1\}$. The pair (W, A) is an ordinary log smooth lc pair with coefficients in the fixed DCC set Λ, and pM_W is nef Cartier. Since D is big and $D_Z \sim_{\mathbb{Q}} D$, the adjoint on the left of (0.24) is big.

For each $1 \leq k \leq d$ let $b_k = b(\Lambda, k, p)$ be the integer in the polarized effective-birationality theorem of Birkar–Zhang. Its conclusion is that $|\lfloor l(K_W + A + M_W)\rfloor|$ is birational for every positive l divisible by b_k when $\dim W = k$, the floor taken on the whole adjoint. Set

$$m = \operatorname{lcm}(p_0, b_1, \ldots, b_d),$$

an integer depending only on d and Φ. Fix a positive multiple l of m and take $k = \dim Z$. For a rational section $v \in \mathbb{C}(W) = K$ of the displayed birational system, pushing its divisor inequality forward using (0.24) gives $\operatorname{Div}_Z(v) + lD_Z \geq 0$. Thus these sections belong to the complete rounded system on Z, and their ratios generate K because the system on W is birational. The complete system on Z is therefore nonempty and generates K, and (0.22) transfers exactly these ratios to X: the common factor ψ^{l/p_0} cancels. The complete source system is nonempty and generates the embedded field $K \subset \mathbb{C}(X)$ for every such l.

The function field of the image of a system's rational map is generated by its section ratios, so each of these maps is birationally equivalent to f. To compare with the Iitaka fibration, choose for the individual pair an integer $r > 0$ such that rL is Cartier. Degrees divisible by $\operatorname{lcm}(m, r)$ form a cofinal divisible subsequence of the Cartier section series of L, and such a subsequence has the same field of section ratios: given u/v in one degree, choose $a \geq 1$ with the multiplied degree in the subsequence and write $u/v = (uv^{a-1})/v^a$. The field defining the Iitaka fibration is therefore K, and $\kappa(L) = \operatorname{trdeg}_{\mathbb{C}} K = \dim Z$. This identifies f with the Iitaka fibration up to birational equivalence. The individual index r is used only for this comparison and does not enter m. □

In words. The moduli denominator theorem says the adjoint on the base can be written with bounded denominators. When the base is reasonably simple, one can do better: a uniform multiple of the adjoint is not merely integral, it is the divisor of an actual rational function. "Reasonably simple" means rationally connected, the algebraic-geometric analogue of connected-by-rational-curves, which forces the linear-algebraic obstruction (the Picard group, and its torsion) to be bounded. The plan is: bound the boundary coefficients and the singularities by global ACC, bound the torsion of the divisor class group by boundedness, then clear denominators and apply the torsion bound.

Technical form.

Proposition 0.5 (Uniform principal multiple over a rationally connected base)**.** *Fix an integer $1 \leq d \leq 4$, a DCC set $I \subset [0,1] \cap \mathbb{Q}$, and a positive integer p. There exists a positive integer $\ell = \ell(d, I, p)$ with the following property. Let Z be a normal projective variety of dimension d with a rationally connected smooth projective resolution. Suppose $(Z, B + M_Z)$ is generalized klt, $B \geq 0$ has coefficients in I, and the b-nef rational b-divisor $\mathbf{M}$ has $p\mathbf{M}$ b-Cartier. If $D = K_Z + B + M_Z \sim_{\mathbb{Q}} 0$, then ℓD is an integral principal divisor. Consequently $a\ell D$ is principal and $h^0(Z, \mathcal{O}_Z(a\ell D)) = 1$ for every integer $a \geq 1$.*

Proof. There are three steps: global ACC gives a finite coefficient set and a uniform positive lower bound for generalized log discrepancies; boundedness up to isomorphism in codimension one bounds torsion in the divisor class group; and clearing the coefficients of the actual adjoint applies that torsion bound.

Technical form. Suppose an exceptional prime E over Z is marked with generalized log discrepancy $a \in (0,1)$. Choose a high projective log resolution $g : W \to Z$ on which the nef data descend and E appears, and write $K_W + B_W + M_W = g^*D$ with $\operatorname{Supp} B_W$ SNC and all coefficients less than one. Arrange also for an effective exceptional Cartier divisor F on W with $-F$ g-ample and $\operatorname{Supp}(B_W) \cup \operatorname{Supp} F$ SNC, by resolving the singular locus and then the marked valuation with centres of codimension at least two so that the negative of each new exceptional divisor is relatively ample.

Choose a small positive rational δ with all coefficients of $B_W + \delta F$ still below one and, when E is marked, $\delta \operatorname{mult}_E F < a$. For a positive ample Cartier A on Z, the divisor $M_W + g^*A - \delta F$ is ample; a general member Λ of a divisible multiple makes $(W, B_W + \delta F + \Lambda)$ sub-klt. There is a principal P on W with $\delta F + \Lambda = M_W + g^*A + P$, and pushing forward gives $K_Z + B + g_*\Lambda = D + A + g_*P$ with $g^*(K_Z + B + g_*\Lambda) = K_W + B_W + \delta F + \Lambda$. So, with $\Xi = B + g_*\Lambda$, the pair (Z, Ξ) is an ordinary effective rational klt pair, and a marked E has ordinary log discrepancy $a - \delta \operatorname{mult}_E F \in (0,1)$.

Apply the extraction corollary of BCHM to the klt pair (Z, Ξ) with the prescribed set $\{E\}$ or the empty set. This gives a projective birational $h : Z^+ \to Z$ with Z^+ $\mathbb{Q}$-factorial and with exactly the prescribed exceptional primes (a small $\mathbb{Q}$-factorial modification in the empty case). Restore the generalized data by $K_{Z^+} + B^+ + M_{Z^+} = h^*D$ with the same nef b-divisor; the pair is generalized klt with rationally trivial adjoint, and B^+ is effective, being the strict transform of B together with $(1-a)E$ in the extraction case.

Now apply global ACC. If the nef trace is not numerically trivial on a model determining $\mathbf{M}$, write $M_W = \frac{1}{p}(pM_W)$ with pM_W nef Cartier, and use the generalized global ACC theorem: the nef weight $1/p$ is a single fixed weight adjoined to the DCC set, so the boundary coefficients lie in a finite set. If instead $M_W \equiv 0$, take a common model $\widetilde{W} \to Z^+$; the trace M_{Z^+} is $\mathbb{Q}$-Cartier and the difference $M_{\widetilde{W}} - \pi^*M_{Z^+}$ is exceptional and numerically trivial over Z^+, so the negativity lemma makes it zero, hence $M_{Z^+} \equiv 0$ and the generalized discrepancies are the ordinary discrepancies of (Z^+, B^+); ordinary global ACC again gives finitely many coefficients. Applying this first to the small modification with no marked divisor shows the coefficients of B lie in a fixed finite set $I_0 \subset I$ and the pairs Z are uniformly generalized ε-lc for some $\varepsilon = \varepsilon(d, I, p) > 0$: otherwise a sequence of marked valuations with discrepancies decreasing to zero would, after extraction, produce boundary coefficients $1 - a_i$ in a DCC set that global ACC would force to be finite, a contradiction.

Technical form. Birkar's boundedness theorem applies to the varieties Z: projectivity, fixed dimension, generalized ε-lc singularities, effective boundary, rationally trivial adjoint, and rational connectedness. It gives, for each Z, a projective V_Z, from one bounded set, isomorphic to Z in codimension one; replacing by finite normalization keeps this and stays bounded. Hence a projective morphism $\mathcal{V} \to S$ over a finite-type base whose normal fibres are isomorphic to the Z in codimension one.

Let $U = Z_{\mathrm{reg}}$. First, $H_1(U(\mathbb{C}), \mathbb{Z})$ is invariant under isomorphism in codimension one of normal varieties: the common open subset, intersected with the regular loci, is each regular locus with a closed subset of complex codimension at least two removed, and removing such a subset from a smooth complex variety does not change the fundamental group (perturb paths and homotopies, relative to boundary, transverse to a smooth stratification whose strata have real codimension at least four). Second, the groups take only finitely many types. Replace S by its reduction and cover by finitely many flat affine strata with $\mathcal{U} = \operatorname{Sm}(\mathcal{V}/S)$ the relative smooth locus; embed $\mathcal{V} \subset \mathbb{P}^N_{\mathbb{C}} \times S$ and identify $\mathbb{P}^N(\mathbb{C})$ with Hermitian matrices $H^2 = H$, $\operatorname{tr} H = 1$. The map $\mathcal{U}(\mathbb{C}) \to S(\mathbb{C})$ is continuous semialgebraic, so the Delfs–Knebusch semialgebraic triviality theorem partitions $S(\mathbb{C})$ into finitely many pieces over which the whole inverse image is a product; each semialgebraic

fibre has the homotopy type of a finite complex. Thus the H_1 take only finitely many finitely generated isomorphism types.

For each Z, $\mathrm{Cl}(Z)$ is finitely generated: a smooth projective rationally connected resolution $r : Y \to Z$ gives a surjection $\mathrm{Pic}(Y) \to \mathrm{Cl}(Z)$ (strict transforms), and since the Albanese map of Y is constant, $\mathrm{Pic}^0(Y) = 0$, so $\mathrm{Pic}(Y)$ is finitely generated by Néron–Severi finiteness. Normality and projectivity give $\Gamma(U, \mathcal{O}_U^*) = \mathbb{C}^*$, and $\mathrm{Pic}(U) = \mathrm{Cl}(Z)$. For every $n \geq 1$ the étale Kummer sequence and classification of finite abelian torsors give

$$\mathrm{Cl}(Z)[n] \simeq H^1_{\text{ét}}(U, \mu_n) \simeq \mathrm{Hom}\big(H_1(U(\mathbb{C}), \mathbb{Z}), \mu_n(\mathbb{C})\big). \tag{0.25}$$

For a fixed Z the orders on the left are bounded because $\mathrm{Cl}(Z)$ is finitely generated; if $H_1(U(\mathbb{C}), \mathbb{Z})$ had a free summand of positive rank the orders on the right would grow without bound, so it is finite. Finitely many finite groups occur, so there is a uniform T divisible by all their exponents, and every torsion class of $\mathrm{Cl}(Z)$ is killed by T: take n its order in (0.25), whose right side has exponent dividing T.

Technical form. Since pM_Z is integral Weil, being the pushforward of a Cartier divisor on a determining model, choose q divisible by p and by all denominators in the finite set I_0; then qD is integral Weil. The relation $D \sim_{\mathbb{Q}} 0$ makes $[qD] \in \mathrm{Cl}(Z)$ torsion: if $nD = \mathrm{Div}(v)$ then $n(qD) = \mathrm{Div}(v^q)$. Hence TqD is principal, and $\ell = Tq$ proves the proposition in dimension d; a common multiple over the finite dimensional range handles the stated subset. All positive multiples of ℓD are principal, and projectivity with integrality gives the one-dimensional section space. □

Finally, the corollary pulling the two theorems together.

Corollary 0.6 (Uniform principal multiple from a fibration)**.** *Fix a finite rational coefficient set $\Phi \subset [0,1] \cap \mathbb{Q}$. There is a positive integer $r = r(\Phi)$ such that the following holds. Let (X, B) be a projective klt fourfold pair with effective boundary having coefficients in Φ, and suppose $K_X + B \sim_{\mathbb{Q}} 0$. If there is a contraction $f : X \to Z$ with $\dim Z > 0$ and with a rationally connected smooth projective resolution of Z, then $r(K_X + B)$ is an integral principal divisor. This includes $B = 0$, and every positive multiple is also principal.*

Plain form. Apply the uniform denominator theorem with zero base class. The base adjoint is rationally trivial, so the rationally connected torsion proposition applies to it, giving a principal multiple. Multiplying the exact presentation by that multiple writes the total multiple as the divisor of an actual function. □

Technical form. Apply Theorem 0.1 to f with the zero divisor as the rational pullback class. It gives uniformly bounded integers $p_0 \mid p$, a function $\psi \in \mathbb{C}(X)^*$, and an exact identity

$$K_X + B + \frac{1}{p_0}\,\mathrm{Div}(\psi) = f^* D_Z, \qquad D_Z = K_Z + B_Z + M_Z, \tag{0.26}$$

with $(Z, B_Z + M_Z)$ generalized klt, B_Z effective with coefficients in a fixed rational DCC set, and $p\mathbf{M}$ b-Cartier with nef data. Since $f^* D_Z \sim_{\mathbb{Q}} 0$ and $f_* \mathcal{O}_X = \mathcal{O}_Z$, the projection formula makes $D_Z \sim_{\mathbb{Q}} 0$.

Proposition 0.5 gives a uniform ℓ and $v \in \mathbb{C}(Z)^*$ with $\ell D_Z = \mathrm{Div}(v)$. Choose r divisible by both ℓ and p, so $p_0 \mid r$. Multiplying (0.26) by r gives

$$r(K_X + B) = \mathrm{Div}\Big((v \circ f)^{r/\ell}\, \psi^{-r/p_0}\Big).$$

Both exponents are integers, so this is the divisor of an actual rational function and includes the integrality of the adjoint multiple. All constants depend only on Φ, since the possible dimensions

of Z form a finite set and the denominator theorem supplies uniform p_0, p and a fixed DCC set. □

The following table collects symbols used in more than one place. A section reference points to the introduction of the symbol.

Symbol	Meaning	Section
(X, B)	projective lc pair of dimension $d \le 4$, boundary B	??
Φ	fixed finite set of allowed boundary coefficients in $[0,1]\cap \mathbb{Q}$	??
$f : X \to Z$	contraction with log Calabi–Yau adjoint	??
$K_X + B \sim_{\mathbb{Q}} f^*D$	rational pullback hypothesis	??
B_Z	discriminant divisor, coefficient $1 - t_P$	??
t_P	log canonical threshold of f^*P over the generic point	??
$\mathbf{M}$, M_W	moduli b-divisor and its trace on a determining model W	??
p_0	generic-fibre trivializing degree, clears Φ	??
p	uniform multiple making $\mathbf{M}$ b-Cartier	??
ψ	principal correction with vertical divisor	??
D_Z	exact base representative, $D_Z = K_Z + B_Z + M_Z$	??
$\mathcal{B}$	DCC set of discriminant coefficients	??
α	coefficient of D_W at a prime $P \subset W$	??
$V_C \to C$	curve slice of a smooth model	??
$F = h^*[c]$	special fibre over the sliced point c	??
Γ, E	dlt boundary and the error $\Gamma - B^c - tF$	??
$N \to C$	endpoint of the relative minimal model program	??
$T = F_{\mathrm{red}}$	reduced special fibre, connected	??
H	effective horizontal boundary with coefficients in Φ	??
(T, B_T)	slc log Calabi–Yau pair obtained by adjunction on T	??
$\mathcal{D}(\Phi)$	DCC set of different coefficients	??
a/m	$p_0\beta$ in lowest terms, the denominator to kill	??
$C' \to C$	cyclic cover $w^m = z$	??
$\pi : Y \to N$	normalized base change, Galois group μ_m	??
s	ambient generator $(\pi^*\theta)^{\otimes p_0}\pi^*\psi\, w^{-a}$	??
u	global generator of $p(K_T + B_T)$	??
r_j	residue ratio on a covering component	??
$m = m(d, \Phi)$	degree generating $\mathbb{C}(Z)$ from sections	??
$\ell = \ell(d, I, p)$	uniform principal multiple on the base	??
$r = r(\Phi)$	uniform principal multiple in the fourfold corollary	??

Conventions. The symbols $\sim$, $\sim_{\mathbb{Q}}$, and $\equiv$ denote linear, rational linear, and numerical equivalence. Tensor products of line bundles are written additively: $A + B$ means $A \otimes B$ and mA means $A^{\otimes m}$. Equalities such as (0.1) are equalities of actual rational divisors for compatible choices of canonical divisors and the displayed rational function; the canonical divisors are fixed using rational top forms and are not rescaled silently. An integral principal multiple rD means $rD = \mathrm{Div}(v)$ for an actual rational function v, which includes integrality of every coefficient of rD. The rounded system of a rational divisor D is the complete system of $\mathcal{O}_X(\lfloor D \rfloor)$; a nonzero v is a section exactly when $\mathrm{Div}(v) + D \ge 0$, and this needs no Cartier condition. The field generated by a system is the subfield of $\mathbb{C}(X)$ generated by ratios of its nonzero sections. The two integers p_0 and p satisfy $p_0 \mid p$; all bounds depend only on the dimension $d \le 4$ and the coefficient set Φ, never on the individual fibration.